

Vision: sending images to Claude

Claude Certified Developer, Foundation (CCDV-F) · UST masterclass

EN

Session 9 of 12 · English notes

Claude is **multimodal** in this course: text, images, and (as named in passing) other media. Vision is not a separate HTTP API. You put an **image content block** next to a text block in the same user message so the model can ground the instruction on pixels.

Image block shape

Earlier labs sent only text blocks. Vision adds a JSON block with `type: image`. Two source styles in class:

- **Base64** — Python reads the file as bytes, encodes, sets `media_type` such as `image/png`, and sends the data. This is what the fire-safety notebook did.
- **URL** — point at an image the API can fetch (e.g. an object in S3 with a reachable URL). You do not embed bytes.

The text block still carries the task (“count the marbles”, “rate fire risk 1–4”). Few-shot or chain-of-thought can live in that text to tell the model *how* to look.

Limits taught in class (verify on docs)

Constraint (20 August lecture)	Figure given	Why it matters
Images per request (across messages)	Up to 100	Insurance packs, contact sheets
File size	5 MB per image	Resize before encode
Max width/height, one image	8000 px	Huge scans
Max width/height, many images	2000 px	Batch photos
Token estimate	width × height / 750 (he also said 250 once — treat as slip)	Vision is not “free”; it consumes the context window

LIMITS MOVE

Formats (JPEG, PNG, GIF, WebP), pixel caps, and token formulae are owned by **vision docs**. If the exam and the transcript fight, the docs win.

You still need a good text prompt

Marble-counting without “they are round glass spheres in this dish” under-counts. The model does not magically segment the photo the way your SME would. Spell out objects, units, and the output schema (one-sentence summary plus a 1–4 rating in the house example).

Worked theme: fire-safety insurance

Several house photos; the user text listed analysis steps (structure, tree-canopy overhang, then assign fire risk 1–4, then one-sentence summary). Changing the PNG changed the rating. Internally Vijay described: bytes → image understanding → embeddings / multimodal transformer path. Your job is media_type + honest pixels + a rubric in text.

Token and privacy hygiene

- Downscale to the smallest size that still shows the defect you care about.
- Do not upload identifiable client property photos to a personal API key.
- URLs must stay reachable for the duration of the request; private buckets need signed URLs or base64 instead.

Exam contrast

Vision = image blocks. PDFs = **document** blocks (next chapter). Do not send a PDF as image/png. Do not send a photo as application/pdf.

Official documentation

- **Vision**

<https://docs.claude.com/en/docs/build-with-claude/vision>

Image blocks, limits, prompting tips.

- **Messages API**

<https://docs.claude.com/en/api/messages>

Content block union (text, image, document).

- **PDF support**

<https://docs.claude.com/en/docs/build-with-claude/pdf-support>

When the file is a document, not a bitmap.

Study notes distilled from the UST CCD Foundation masterclass. Prefer Anthropic's current docs over session anecdotes. Not official exam material.